Stellar Magnetism Through the Instability Strip

James Barron

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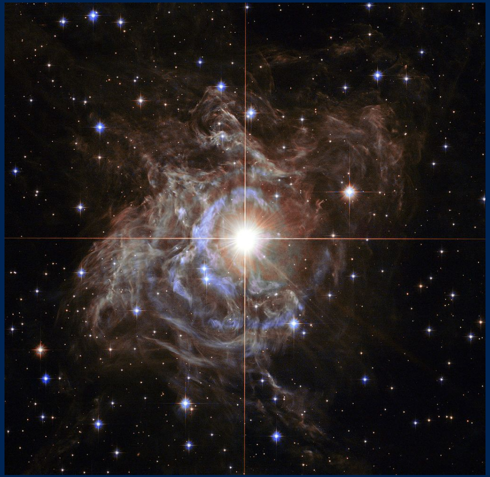
October 26, 2023

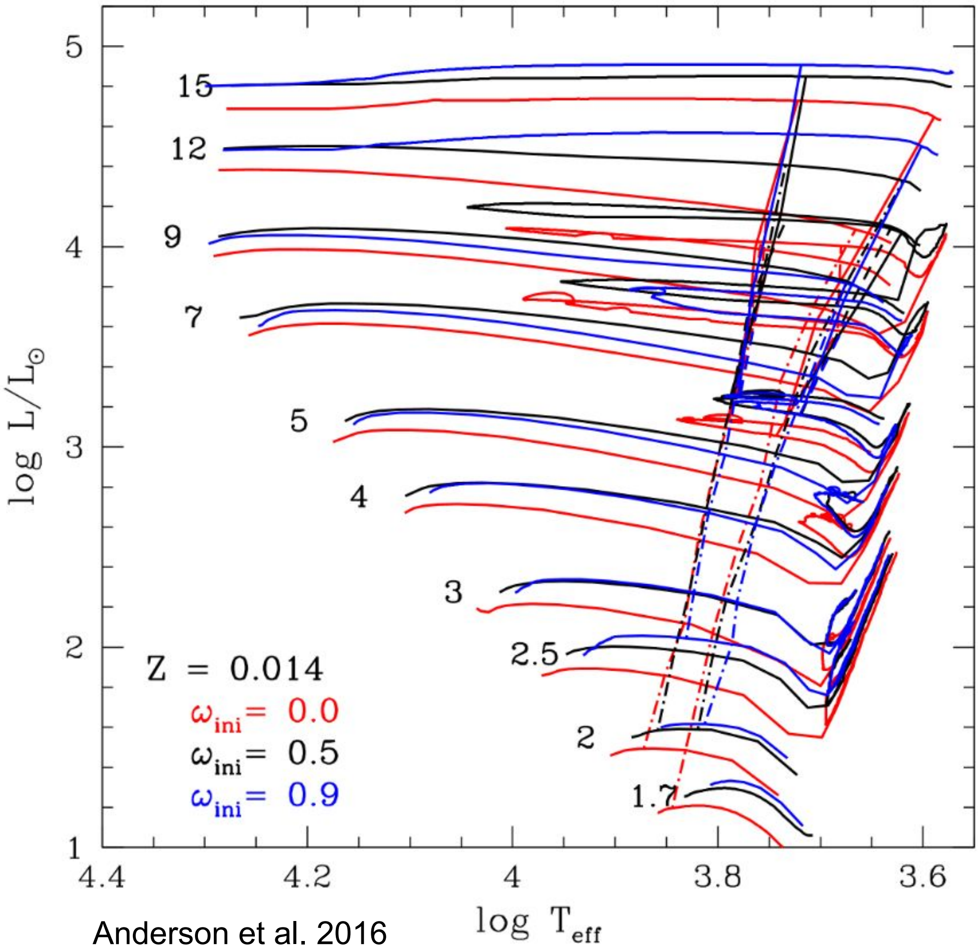


RS Puppis

Queen's University

Introduction: Classical Cepheids

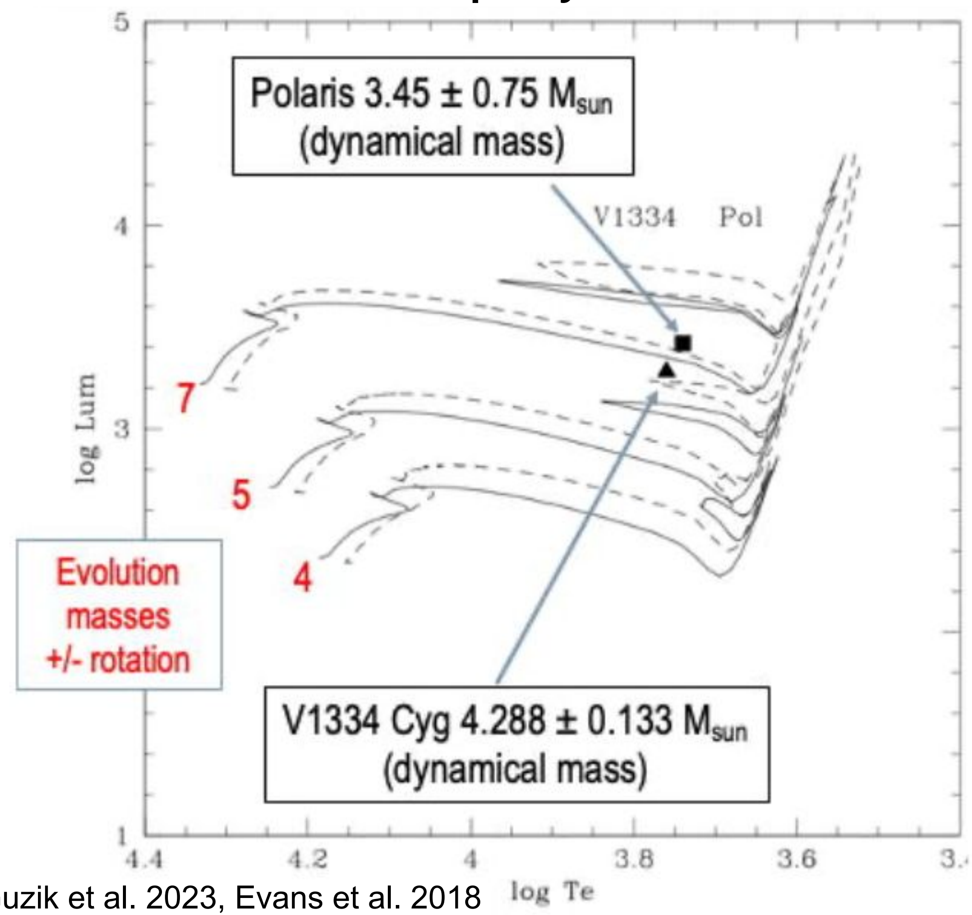
1. Young Pop-I Yellow Supergiants.
2. Intermediate to High Mass (4 - 15 M☉).
3. Luminous (10^3 - 10^5 L☉).
4. Temperature ~ 4500 - 7000 K
5. Periods of ~2 - 60 days.
6. Subclass of s-Cepheids, low amplitude, overtone pulsators.



Anderson et al. 2016

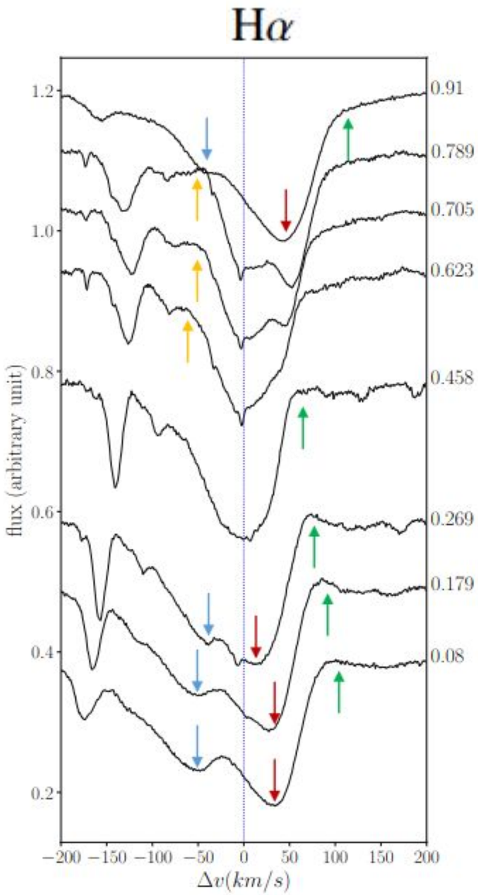
Introduction: Outstanding Questions

Mass Discrepancy Problem

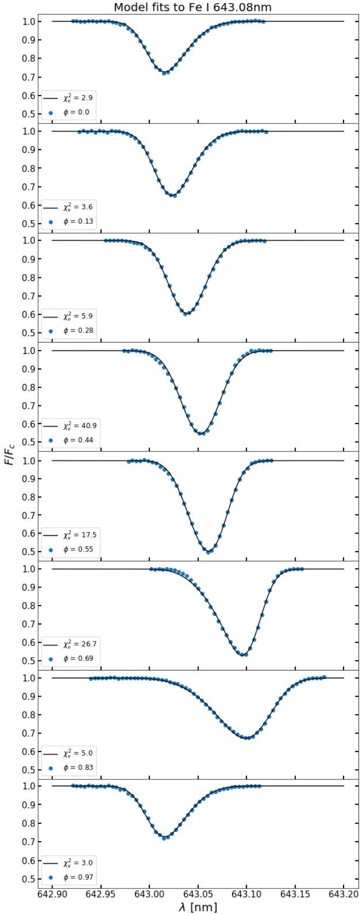


Guzik et al. 2023, Evans et al. 2018

Atmospheric Dynamics and Line Profile Modelling



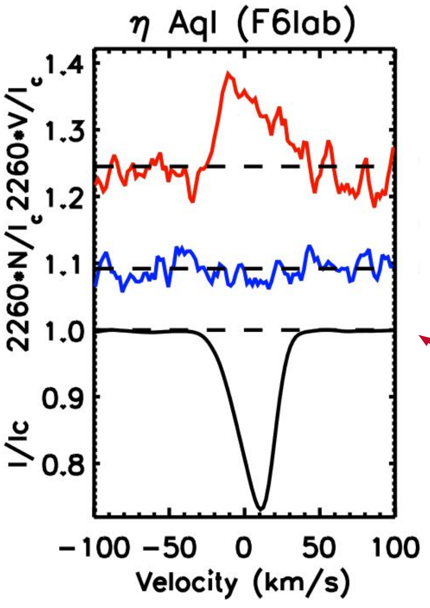
Hocde et al. 2020



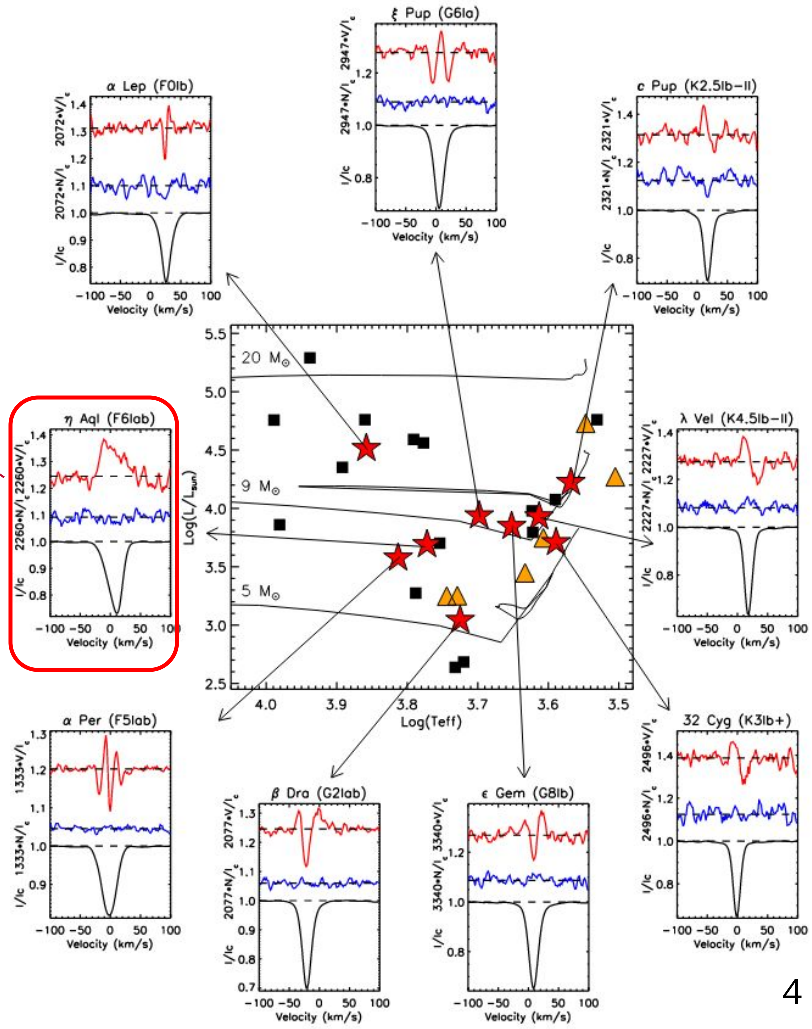
Introduction: Motivation

- Previously η Aql. was the only Cepheid with a magnetic detection in high resolution spectropolarimetry.

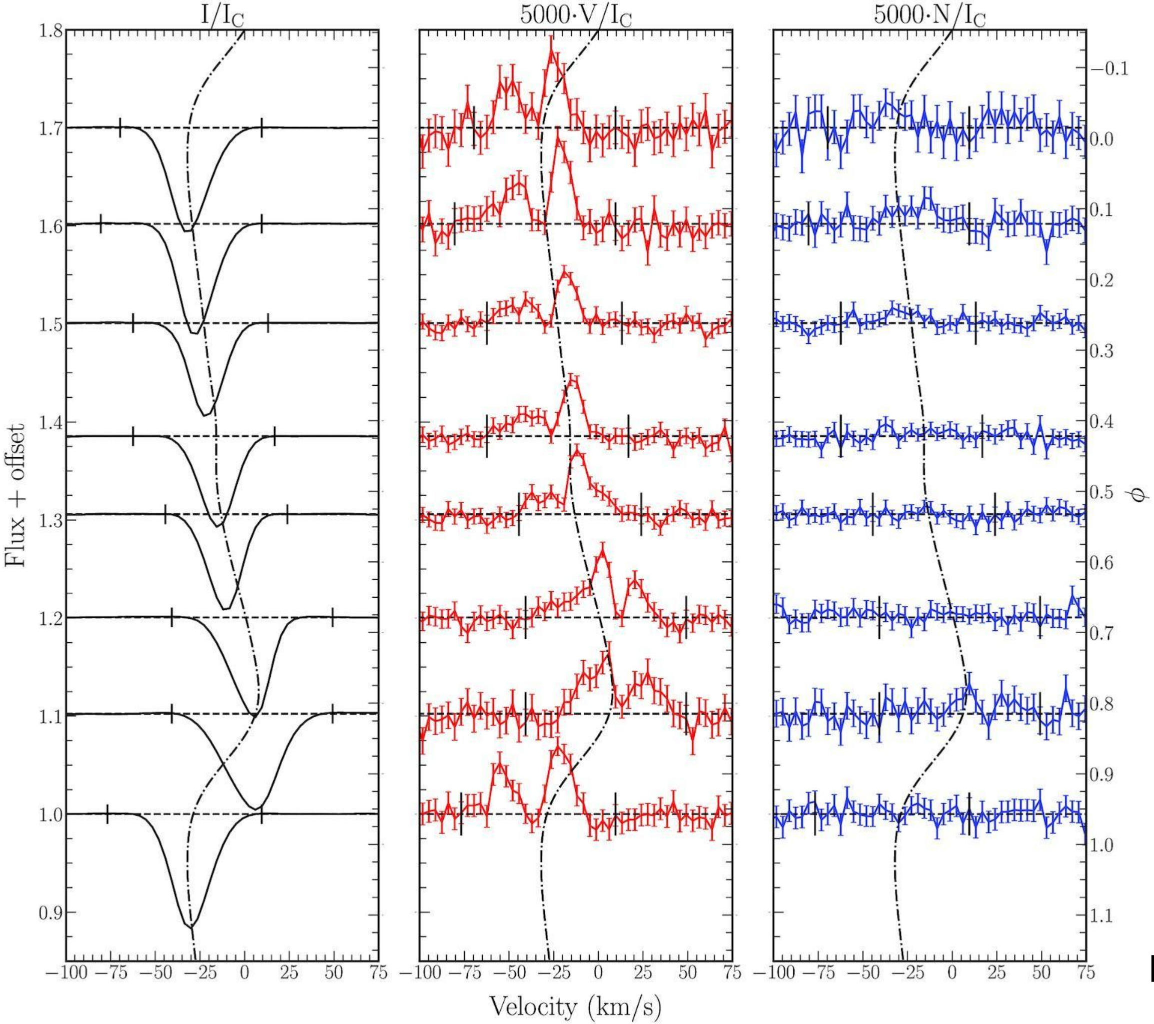
- LSD Stokes *V* profile is unusual, a single asymmetric lobe.



Grunhut et al. 2010



η Aql



Barron et al. 2022

First Magnitude Limited Spectropolarimetric Survey

Motivating Questions

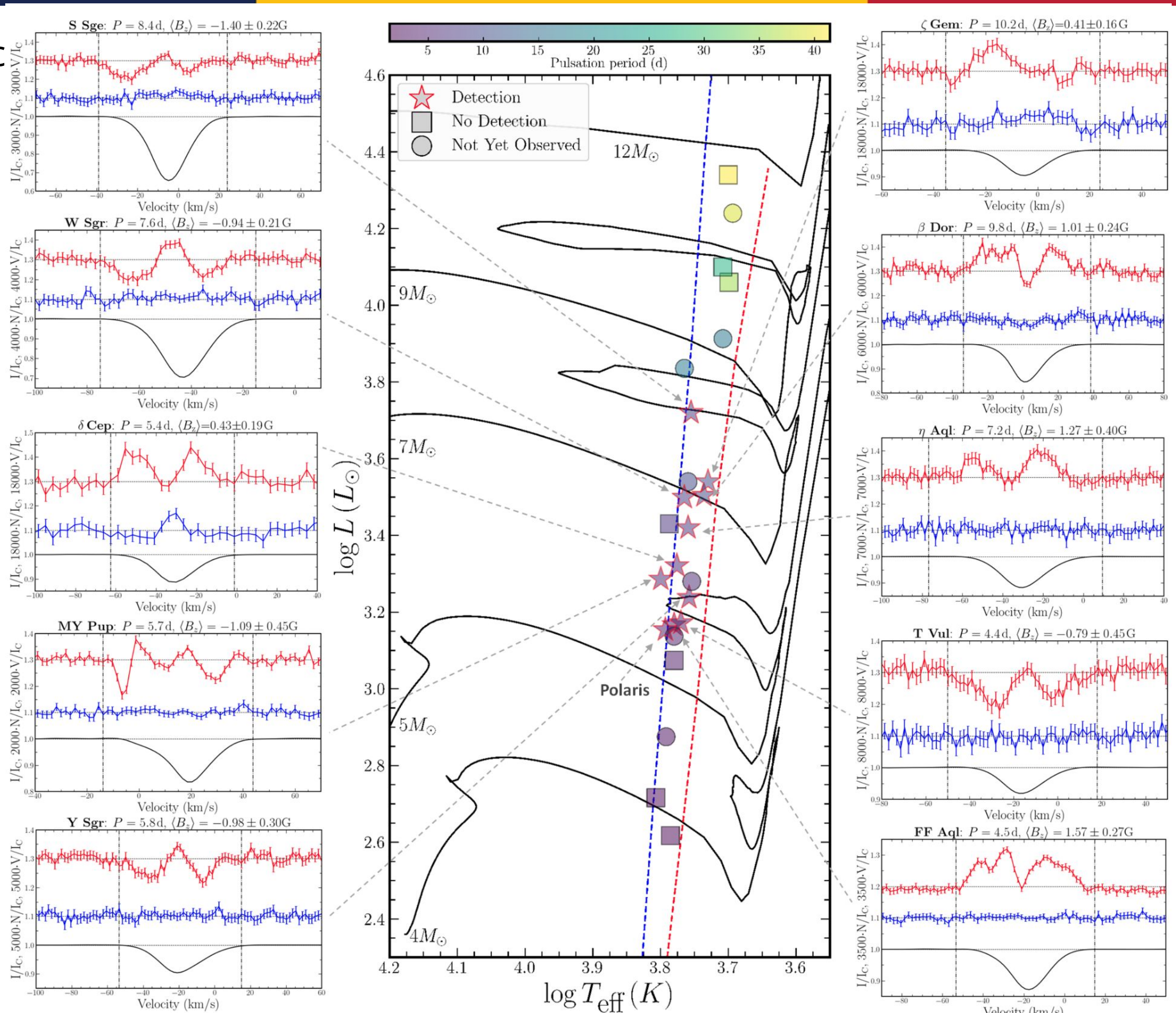
1. Are Cepheid photospheric magnetic fields generally detectable?
2. What types of longitudinal magnetic field strengths and Stokes *V* morphologies do Cepheids possess?

Survey

- Magnitude limited sample (~25 targets) of the brightest ($V_{max}<6$ mag) classical Cepheids.
- Twenty CCs observed to date with ESPaDOnS and HARPSpol.
- Require high S/N of ~4500 at 500 nm (CFHT).
- Long charged times, up to 5 hours per target.
- Aim to observe at maximum brightness.

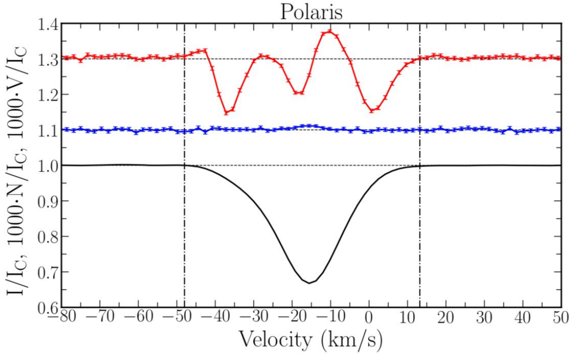
Twelve new magnetic detections.

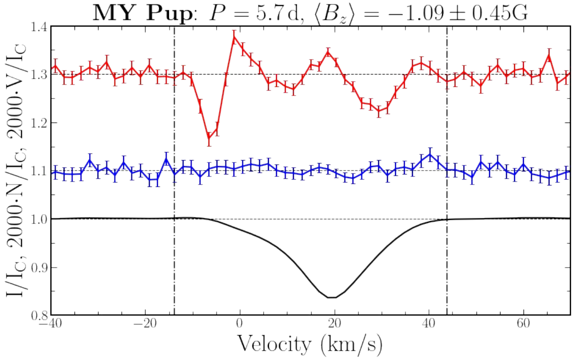
Unexpected variety of LSD Stokes *V* morphologies.



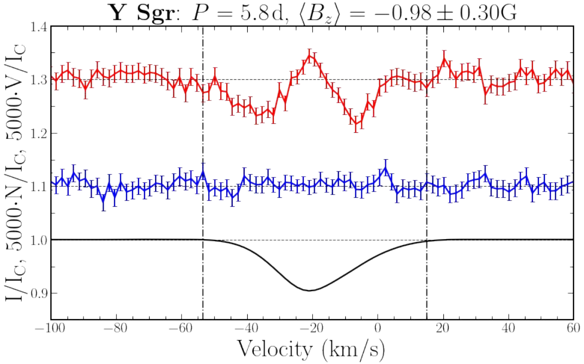
Origin of LSD Stokes *V* Asymmetry: s-Cepheids vs. Fundamental

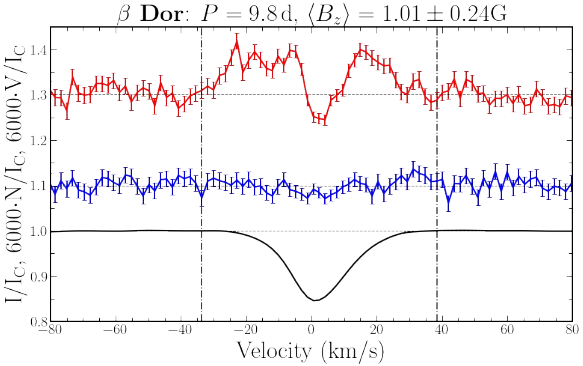
Low Amplitude s-Cepheids

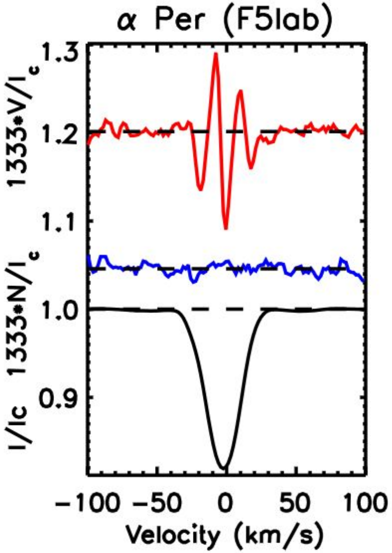




Fundamental Cepheids



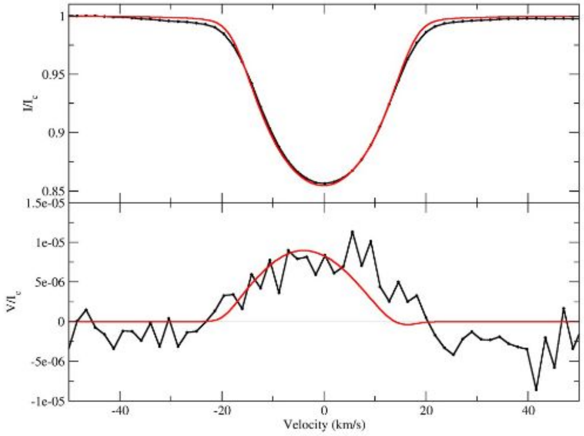


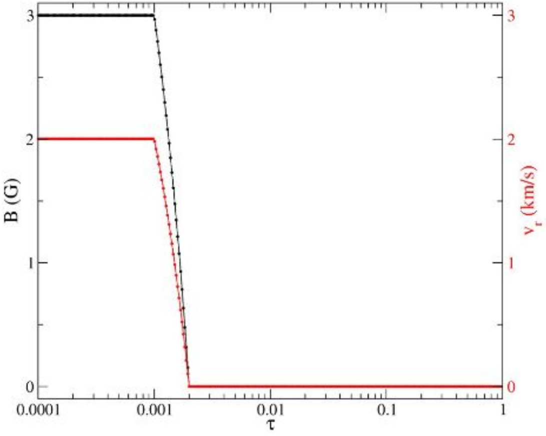


Grunhut et al. 2010

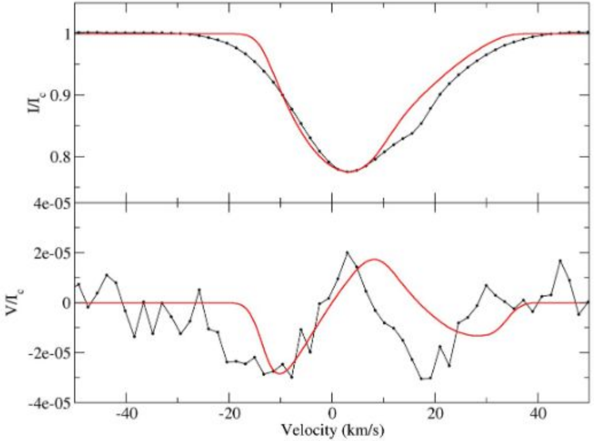
Origin of LSD Stokes *V* Asymmetry: Modelling Prospects

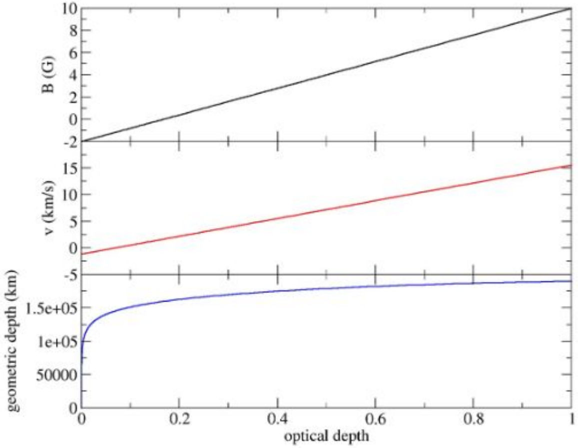
Sirius





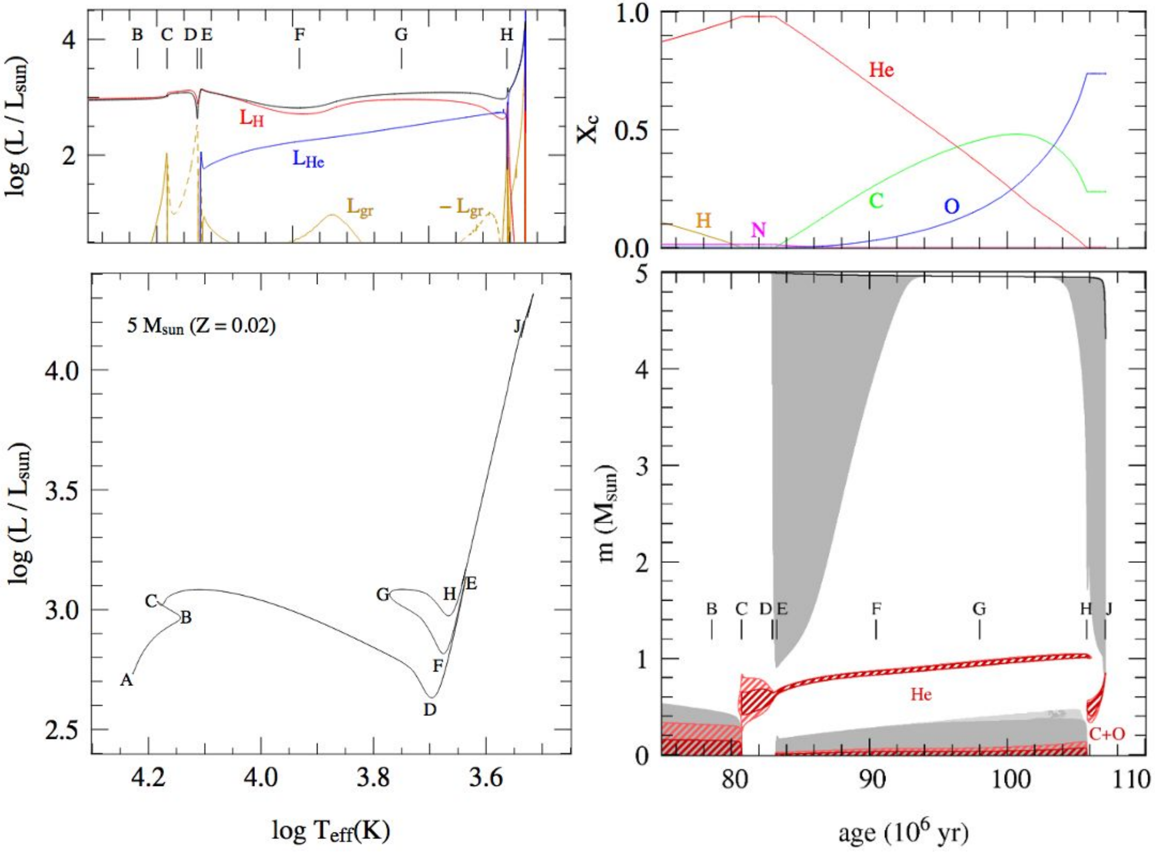
Y Sgr



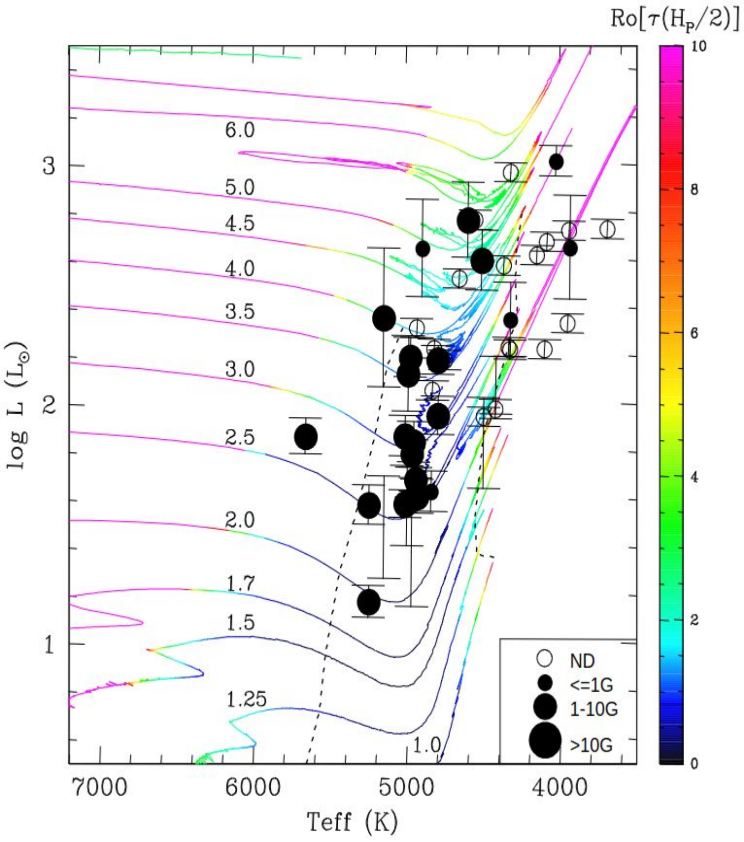


Courtesy of Colin Folsom

What is the origin of Cepheid magnetics fields?

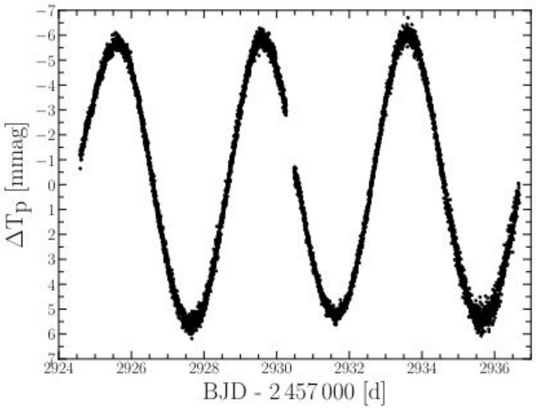


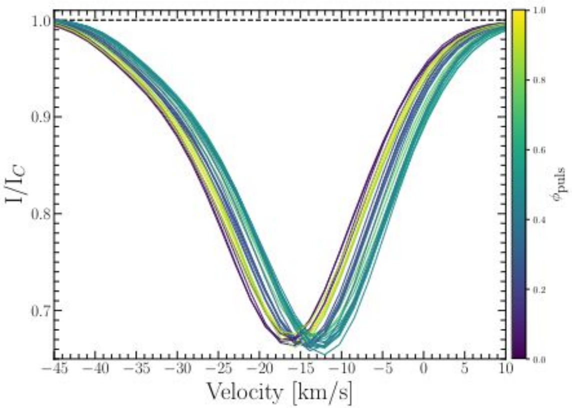
Onno Pols Lecture Notes



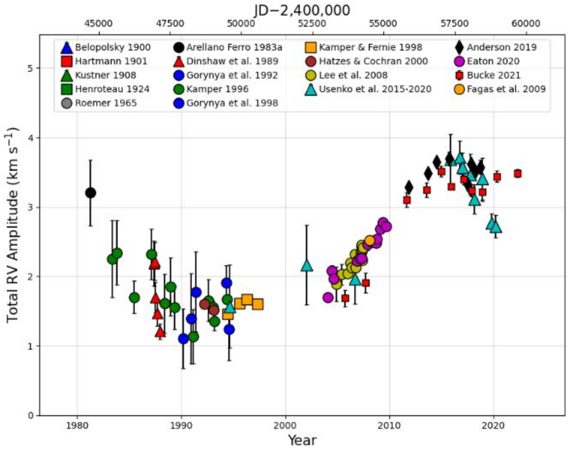
Charbonnel et al. 2017

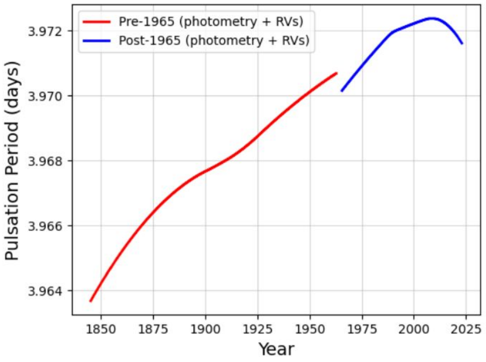
A Dedicated Magnetic Study of Polaris





xkcd.com

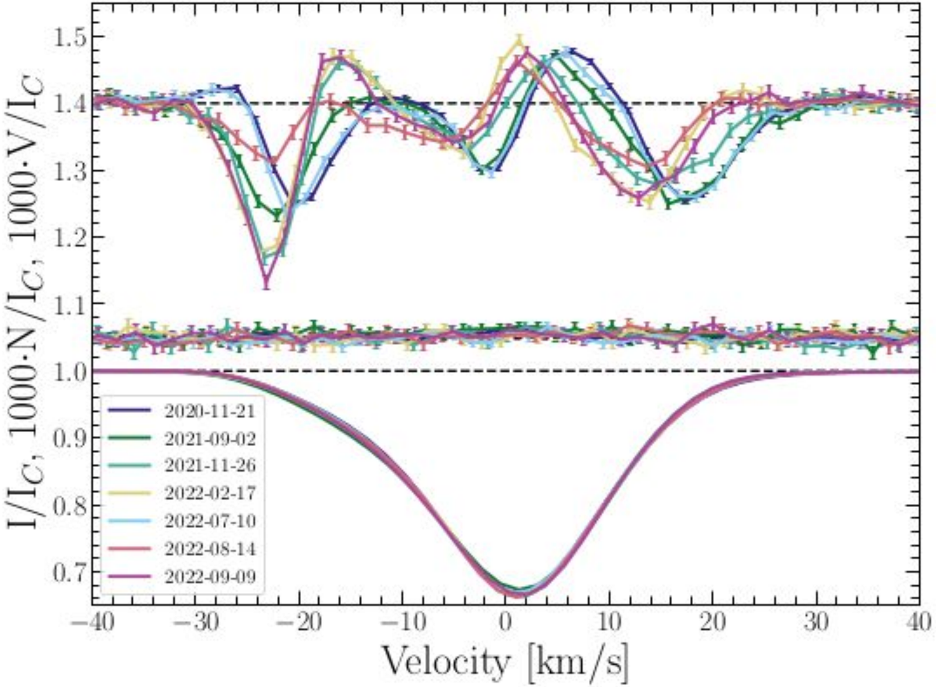


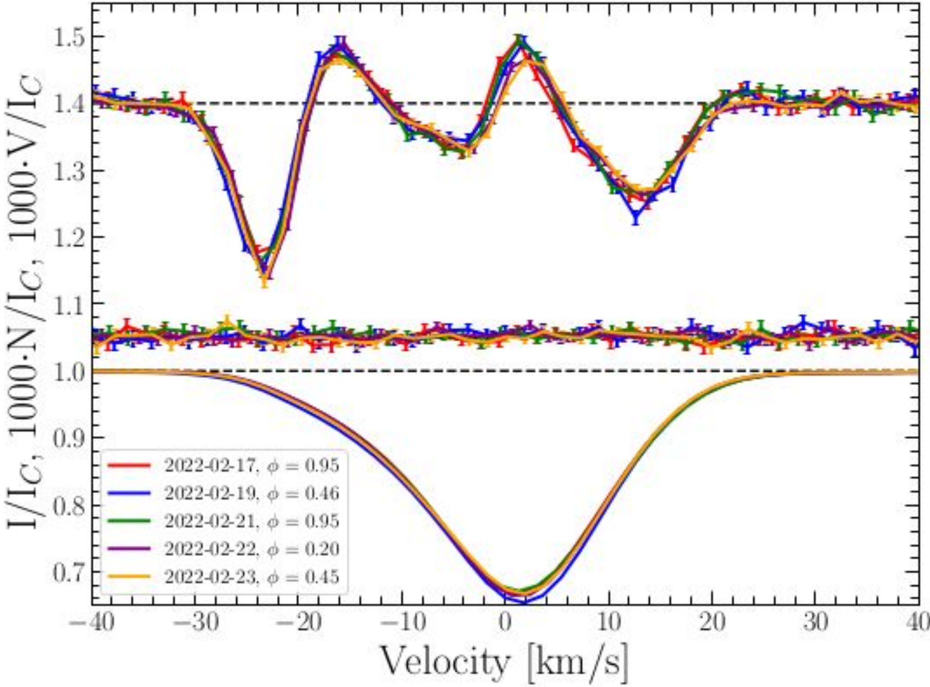


Torres 2023

A Dedicated Magnetic Study of Polaris

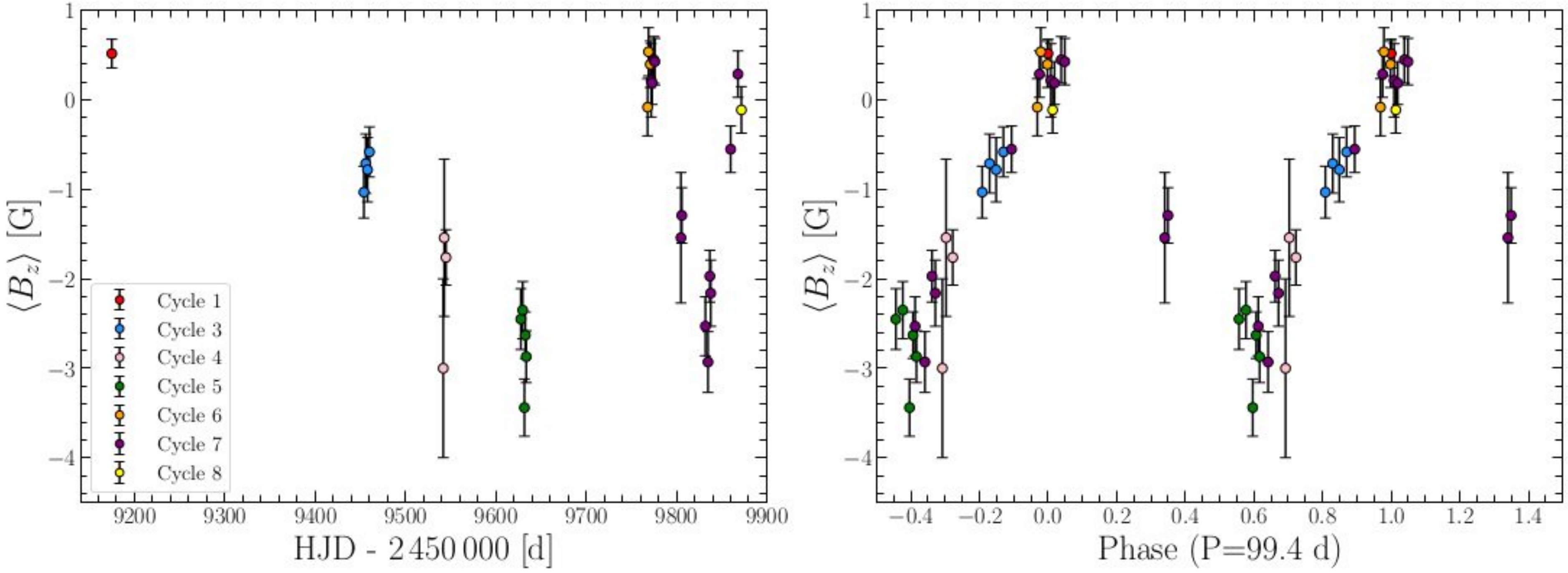
- 34 observations obtained over the last 3 years, will conclude this semester.
- Aim to perform first direct measurement of the rotation period of a CC.
- Explore feasibility of Stokes V modelling and ZDI.



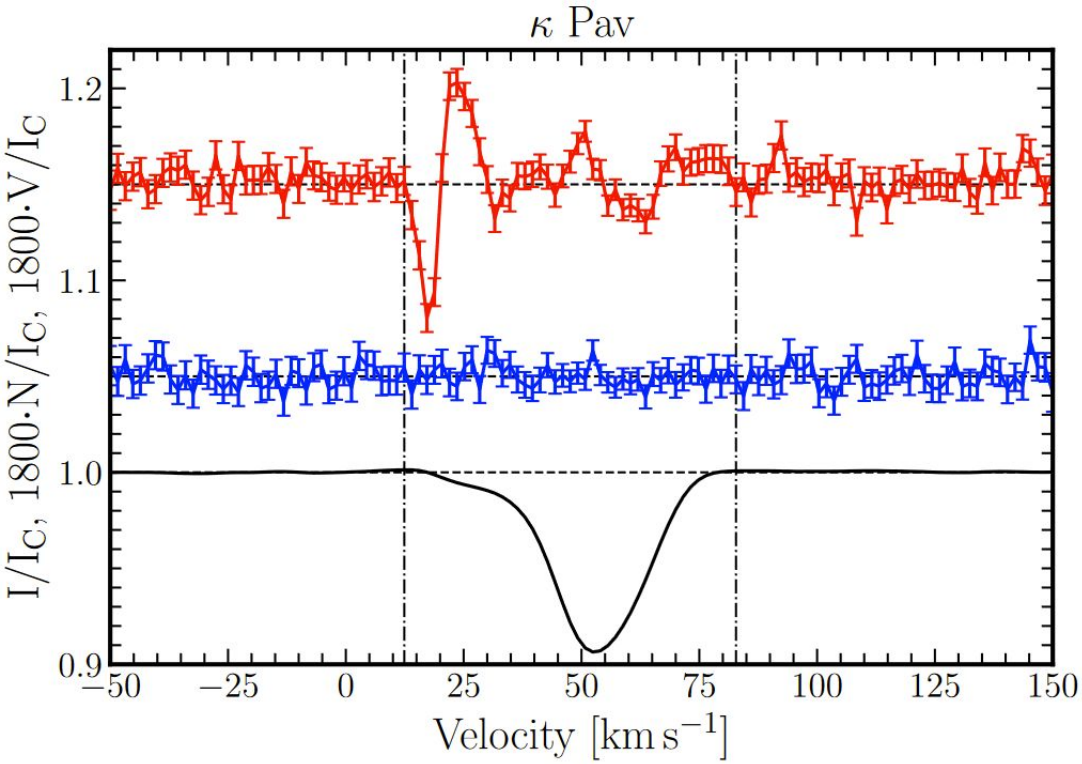


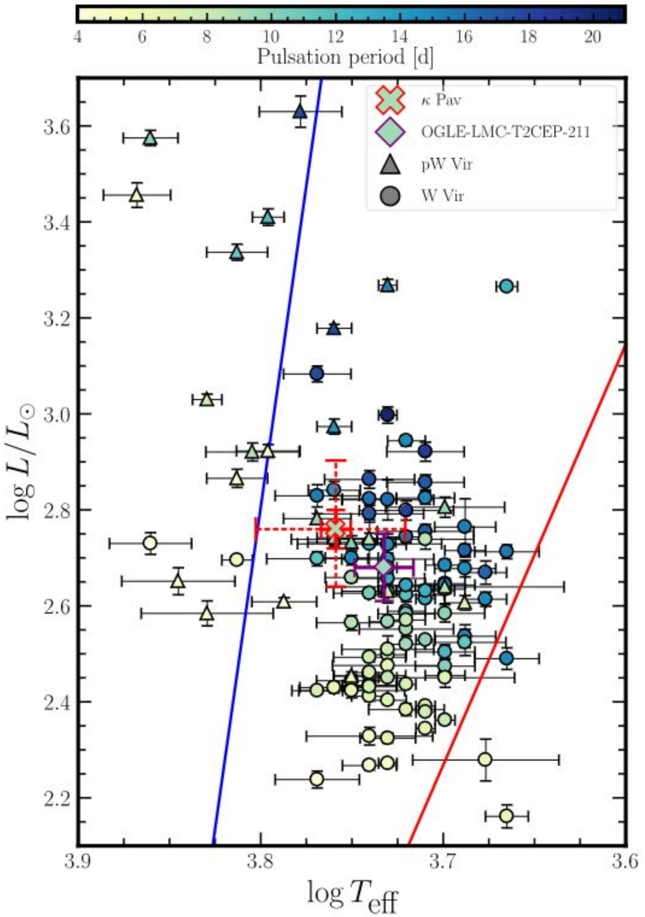
A Dedicated Magnetic Study of Polaris

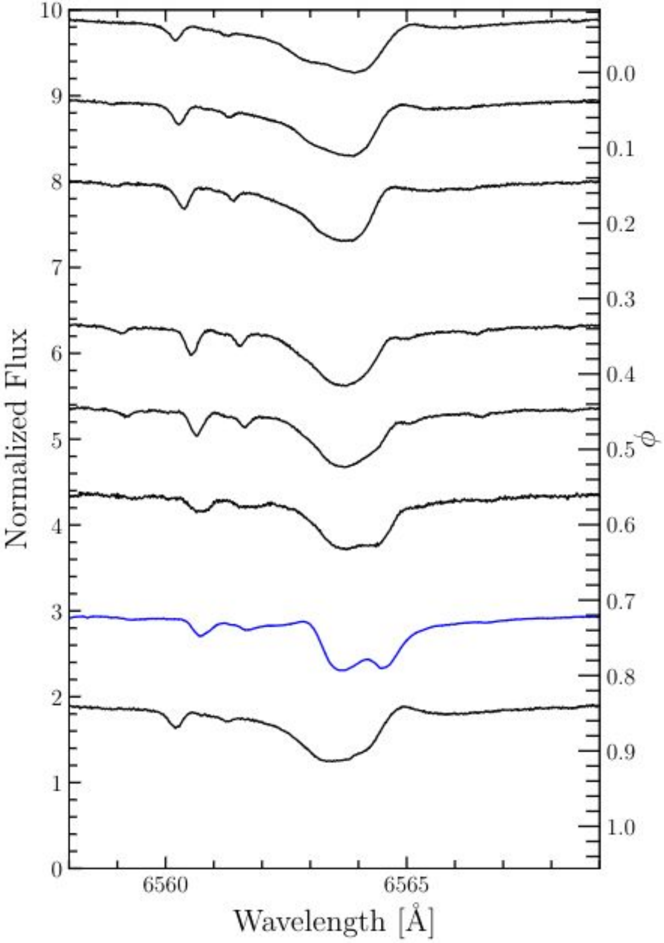
- Tentative rotation period of 99.4 days.
- If confirmed this would make Polaris essentially a solved system.

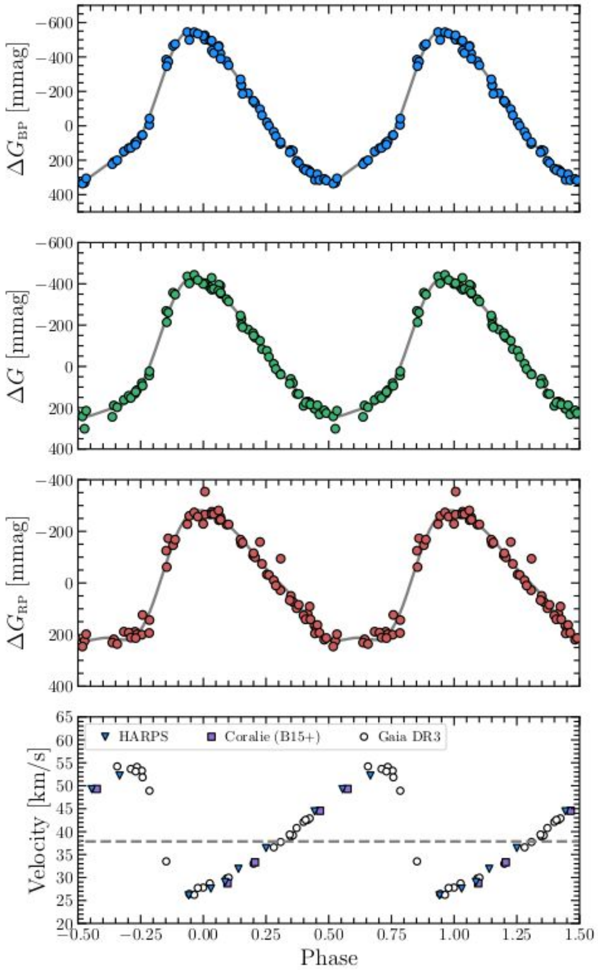


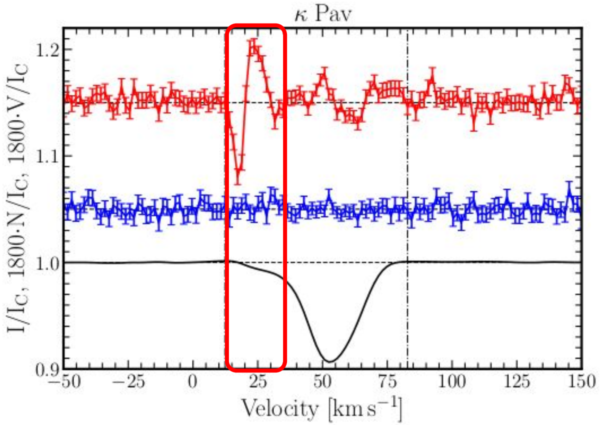
First Magnetic Detection in a Type II W Vir Cepheid: κ Pav



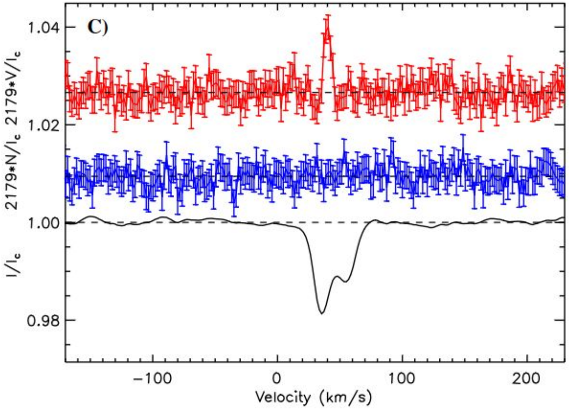








Post AGB Star R Scu



Sabin et al. 2015

Conclusions

- Cepheid magnetic fields can be regularly detected when observed with sufficient precision.
- Longitudinal field strengths of ~1 G.
- LSD Stokes V profiles are asymmetric, and show a diversity of morphologies (single lobed, double lobed, complex).
- Hypothesize that the Stokes *V* morphologies are the result of atmospheric velocity / magnetic field gradients and possibly shocks.
- Circular polarization may provide a new probe of Cepheid atmospheric dynamics.
- Magnetic monitoring of Polaris will provide the first direct measurement of a Cepheid rotation period.